

JONATHAN EARP

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First-year Electrical and Mechanical Engineering student. I build and fly my own drones, have worked on a sensor research project at Lawrence Berkeley National Laboratory since 2024, and captained a VEX robotics team to a state title and a Worlds division final. I'm applying to the Electromagnets, Dynamics and Static technical teams. My closest experience is brushless motors and ESCs, IMU data and PID tuning, and building robot mechanisms.

EDUCATION

University of Edinburgh Edinburgh, UK · 2026 – 2031
MEng Electrical and Mechanical Engineering, Year 1

The Head-Royce School Oakland, CA · Jun 2026
US high school diploma. GPA 3.80/4.00 unweighted. Honours Calculus, Physics, Chemistry, Data Science

ENGINEERING EXPERIENCE

Lawrence Berkeley National Laboratory — Student Research Assistant Berkeley, CA · Apr 2024 – Present
StoveCam: low-cost camera sensor for an indoor air quality and childhood asthma study

- Built a Raspberry Pi 5 unit with RGB and thermal cameras that captures the cooktop every 60 s in occupied homes, then reworked the mounts, wiring and capture scripts until it ran reliably across different kitchens.
- Wrote a three-stage Python labeling pipeline: an on-device brightness gate, a low-cost cooking screen, then Gemini vision for food, method, burner and cookware. 40,000+ RGB/thermal pairs from three homes at \$0.018 per analyzed pair.
- Validated the output against 506 hand-graded pairs: mean score 8.1/10, with zero burner or cooking-method errors. Most failures traced to residual-heat false positives. Abstract submitted to ISES 2026.

University of Pennsylvania, Singh Center for Nanotechnology — ESAP Participant Philadelphia, PA · Summer 2025

- Class 100 (ISO 5) cleanroom: photolithography, microfluidics, CdSe quantum dot synthesis, SEM and AFM imaging.
- Built an adaptive audio control device with Prof. Gyuseok Kim and graduate TAs, from first prototype to working demo.

PROJECTS AND TEAMS

Head-Royce VEX Robotics — Team Captain Oakland, CA · 2022 – 2026

- Designed and built robot mechanisms, then tuned PID control with the programmers over repeated test runs.
- Kept the team building through a workspace move with a set schedule and clear ownership of each subsystem.
- State Champion, Worlds Division Finalist, 2× Tournament Champion, 2× Excellence, 3× Build, 2× Amaze, 1× Design.

FPV and UAS Builds — Personal | Betaflight, ArduPilot, ExpressLRS 2023 – Present

- Built 8 multirotors from 45 mm to 5 in. Selected the motors, STM32-based ESCs and props, then soldered and integrated the flight controller, receiver, GPS and video system on each.
- Calibrate GPS, compass, barometer and IMU on every build, and tune PID loops over repeated test flights using flight logs.
- 400+ flight hours, including car-chase, real estate and music video shoots. FAA Part 107 certified (US commercial drone licence). All crash repairs and rebuilds done myself.

3D Reconstruction from FPV Flight Data — S.P.L.O.O.G.E | Python, COLMAP, Rust, WebGL Aug 2026

- Built a pipeline from raw DJI O4 Pro video and gyro telemetry to one 3D Gaussian Splatting model of two flights six days apart: 7,470 of 7,471 frames registered at 0.82 px mean reprojection error.
- Estimated the uncalibrated lens focal length by regressing optical flow against gyro rotation ($R^2 = 0.967$), and cut video-to-render misalignment from 72.3 px to 0.89 px by undistorting the footage.
- Traced a rendering defect to the viewer's near-plane cull after measuring four other fixes, including a depth-distortion loss I implemented in Rust. Holes fell from 51.2% to 4.4%, with no regressions across 56 test poses.

Automated Video Rendering Pipeline — P.E.T.E.R | Python, OpenCV, Playwright, ffmpeg 2026

- Wrote a 4,100-line Python pipeline that renders video clips as Desmos equations and uploads them. 4.4M plays.
- Profiled rendering (94% capture, 4% tracing) and parallelized capture for a 1.55× speedup, then found a timing bug corrupting 39 of 40 frames and fixed it by sweeping the wait threshold until output converged.

SKILLS

Electrical & avionics: Brushless motors, STM32 ESCs, soldering, Betaflight, ArduPilot, ExpressLRS, analog and DJI video

Controls & sensors: PID tuning, IMU and gyro data, quaternion orientation, sensor calibration, flight-log analysis

CAD & fabrication: Onshape, Fusion 360, 3D printing, CNC, laser cutting, bandsaw, drill press, photolithography

Programming: Python (NumPy, OpenCV, Playwright), Rust, COLMAP, JavaScript, WebGL, Arduino, Raspberry Pi / Linux, Git

LEADERSHIP AND OUTREACH

- Medical Advocacy, President.** Ran a blood drive that brought in 120 donors, and set up a partnership with a local hospital.
- Eagle Scout, Senior Patrol Leader.** Taught a month-long robotics class at an elementary school for my Eagle project, and left the curriculum behind so someone else could run it again.